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CHAITANYA **KANDEKAR**

RESEARCH PROFILE

Computational mechanics researcher specializing in phase-field fracture modeling and multi-physics coupling. Developed partitioned simulation frameworks for electro-chemo-mechanical and thermo-mechanical fracture using preCICE and finite element methods. Expertise in nonlinear FEM, solver coupling, and high-performance scientific computing with applications to stress corrosion cracking and fire-induced failure.

RESEARCH EXPERIENCE

Jun 2021 - present **Wissenschaftlicher Mitarbeiter (Research Associate)** · Helmut-Schmidt University / University of the Federal Armed Forces Hamburg

- Project KIBIDZ - Thermo-hydro-mechanical fracture modelling and simulations for concrete under fire loads
- Project ECoMaD - Electro-chemo-mechanical fracture modelling and simulation for stress corrosion cracking
- Conducted tutorial lectures on continuum mechanics to master students, and MATLAB programming tutorials for bachelor students

Apr 2019 - May 2021 **Wissenschaftlicher Hilfskraft (Research Assistant)** · Leibniz University Hannover

- Inst. of Continuum Mechanics (18 months)- Multi-scale simulations of 3 dimensional fracture using phase field in MATLAB, Nozzle simulations
- Inst. of Dynamics and Vibration (6 months)- Fluid simulations of lubrication layer of the journal bearing in MATLAB
- Inst. of Mechanics and Computational Mechanics (3 months)- Stochastic simulation of a beam using random field in MATLAB

Sep 2016 - Mar 2018 **Design Engineer** · Infinarc Engineering LLP, Pune

- Mechanical Design, Calculation, 3D Modelling, Drafting, Structural and Dynamic Simulations
- Product Development, Material Selection and Procurement, Fabrication, Assembly and Maintenance
- Successfully designed and fabricated various machines such as Autonomous Pesticide Spraying Vehicle, Adhesive Dispensing Machine, Tube beading machine, Phosphating plant set up

EDUCATION

Jun 2021 - present	**Dr.-Ing. - Computational Engineering** Grade: 1.0 (German scale - excellent) · Helmut-Schmidt University / University of the Federal Armed Forces Hamburg **Thesis:** Multi-physics and multi-scale computational frameworks for environment-induced phase-field fracture
Apr 2018 - May 2021	**Master - Computational Methods in Engineering** Grade: 1.5 (German scale - very good) · Leibniz university, Hannover-Germany **Thesis:** A Phase Field model for electro-chemo-mechanically induced fracture at the electrode-catalyst interface of Polymer Electrolyte Water Electrolysis cell
Jun 2012 - Jun 2016	**Bachelor - Mechanical Engineering** Grade: 2.2 (German scale - good) · Savitribai phule university of pune, Pune-India **Thesis:** Improving Plant Capacity and productivity along with automation in filter line

LANGUAGES

English (C1)
German (B2)
Hindi (native)
Marathi (native)

TECHNICAL SKILLS

Finite Elemental Packages - FEniCS, preCICE, Abaqus(UMAT), ANSYS
Programming Languages - MATLAB, Python, Fortran, C
CAD modeling Packages - CATIA, Solidworks
Pre/Post-processing Packages - Gmsh, Paraview, Hypermesh
Utility Packages - Microsoft Office, Latex, Github

PUBLICATIONS

Feb 2025	An approach for a novel multiscale partitioned framework for predicting collapse of buildings under fire loads, 12th International Conference on Fracture Mechanics of Concrete and Concrete Structures (***FraMCoS***-12). https://framcos.org/FraMCoS-12/Full-Papers/1319.pdf
Jun 2024	Mastering the complex time-scale interaction during Stress Corrosion Cracking phenomena through an advanced coupling scheme, Computer Methods in Applied Mechanics and Engineering (***CMAME***), Volume 428. https://doi.org/10.1016/j.cma.2024.117101
April 2024	Crack propagation in anisotropic brittle materials: From a phase-field model to a shape optimization approach, ***Engineering Fracture Mechanics***, Volume 303. https://doi.org/10.1016/j.engfracmech.2024.110065
Sep 2023	Toward coupled fire-structure simulations for forcasting smoke leakage in case of concrete under fire, Proceedings in Applied Mathematics and Mechanics (***PAMM***), Volume 23. https://doi.org/10.1002/pamm.202300250
Mar 2023	A partitioned computational framework for damage evolution in stress corrosion cracking utilizing phase-field, Proceedings in Applied Mathematics and Mechanics (***PAMM***), Volume 22. https://doi.org/10.1002/pamm.202200211
Oct 2022	Electro-Chemo-Mechanical Induced Fracture Modeling in Proton Exchange Membrane Water Electrolysis for Sustainable Hydrogen Productiong, Computer Methods in Applied Mechanics and Engineering (***CMAME***), Volume 400, 1 October 2022. https://doi.org/10.1016/j.cma.2022.115580
Feb 2017	User Authorized Ignition System, International Journal of Automobile Engineering Research and Development (***IJAuERD***), ISSN (print):2277-4785, VOL 7, ISSUE 1, EDITION: FEB 2017.

CONFERENCE PRESENTATIONS

Sep 2025	Computational framework for coupled simulations of concrete structures under fire loads, Baustatik und Baupraxis Forschungskolloquium (***IBB*** 2025), Bad Ball, Germany
Sep 2025	An approach for a novel multiscale partitioned framework for predicting collapse of buildings under fire loads, ECCOMAS International Conference on Computational Plasticity (***COMPLAS*** 2025), Barcelona, Spain
July 2025	A partitioned computational framework for exploiting damage in stress corrosion cracking of metal matrix composite using phase-field method, International Conference on Damage Mechanics (***ICDM*** 2025), Singapore, Singapore
July 2025	On a hybrid simulation using a phase-field approach for predicting the concrete behavior under fire, European Solid Mechanics Conference (***ESMC*** 2025), Lyon, France
Sep 2024	Film rupture-dissolution-repassivation phase-field model to exploit damage evolution in stress corrosion cracking: a partitioned computational framework, Material Science and Engineering (***MSE*** 2024), Darmstadt, Germany
Sep 2023	An advanced multi-phase-field approach for coupled electro-chemo-mechanical problems using the preCICE library applied to fcc mirocstructure, ECCOMAS International Conference on Computational Plasticity (***COMPLAS*** 2023), Barcelona, Spain
Aug 2023	An advanced multi-phase-field approach for electro-checmo-mechanical dissolution driven SCC, European Corrosion conference (***EUROCORR*** 2023), Brussels, Belgium
Jun 2023	An advanced coupling approach for Solving Corrosion and Fracture Mechanics using preCICE, ECCOMAS Internationation conference for Coupled Problems (***COUPLED*** 2023), Chania, Greece
Aug 2022	Multi-phase-field framework for damage evolution in stress corrosion cracking, European Corrosion conference (***EUROCORR*** 2022), Berlin, Germany
Aug 2022	Computational framework for electrochemical dissolution driven stress corrosion cracking utilising a multi- phase-field approach, 92nd International Association of Applied Mathematics and Mechanics (***GAMM*** 2022), Aachen, Germany
Feb 2022	A coupled multifield modelling approach to describe material behaviour of concrete under fire load, 147th Nord-deutsche Mechanik Kolloqium (***NDMK*** 2022), Braunshweig, Germany

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EXTRA-CURRICULAR PROJECTS

Sep 2018- Sep 2019 **Formula Student Competitions 2019** · FSN, FSA, FSS

- Team - 'Horsepower Hannover' participating for Netherlands, Austria and Spain
- International competitions for designing and building formula prototype electric cars
- Working in the chassis department for designing and testing Carbon Fibre sandwiches for monocoque chassis and also responsible for Structural Equivalency Spreadsheet

Sep 2014- Jul 2015 **Formula Student Competitions 2015** · SAE SUPRA

- Team - 'The Metal Falcons' participated for national competition in India
- All India Rank – 9th out of 167 teams
- Led the team as a Steering system department and Innovation department head. Worked in the design, analysis and fabrication of steering system. Presented the 'Cost Analysis Report

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HONOURS & AWARDS

Jul 2015– May 2016	Elected as the student council member – '**Technical Secretary**' and represented 6000 students, the university, and carried out 26 technical activities and events
Dec 2015	**Campus Ambassador** - represented Engineering Cluster Pune and D Y Patil College, Pune
Dec 2014	**Campus Ambassador** for SGT International- organised 'Auto Factory Bikers Workshop' by IIT-BHU